

The Boundary of Graph Sparsification for Community Detection

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Abstract—

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I. Introduction

Community detection on large graphs is bounded by edges, not by nodes. A social network with a hundred million accounts may carry ten billion links, and every algorithm that partitions it—modularity optimization, flow simulation, label propagation, spectral bisection—pays for those links on every pass. The nodes cannot be discarded, because a community assignment is required for each of them; sampling vertices changes the problem rather than shrinking it. Edges are different. If most of them are redundant for the purpose of recovering community structure, then removing them before detection would buy speed for nothing. That possibility has motivated fifteen years of work on sparsify-then-detect pipelines, and the methods that implement it now ship in widely used graph libraries and in at least one production system.

The founding answer is affirmative, and it is more careful than it is usually remembered. Satuluri, Parthasarathy and Ruan [1] introduced local similarity sparsification, in which every node retains the incident edges whose endpoints share the most neighbours, and reported speedups “often in the range 10–50, with little or no deterioration in the quality of the resulting clusters,” adding that “for at least two of the four clustering algorithms, our sparsification consistently enables higher clustering accuracies.” Every clause of that sentence is qualified. The two algorithms are fixed- k , balance-constrained cut partitioners; the accuracy is agreement with external ground-truth communities, not any objective computed on the graph; the speedups are measured against clustering runs that take hours; and the graphs on which the gains are largest are dense—average degrees of 65, 76 and 94—and, in two cases, known to contain a substantial fraction of spurious edges. The authors also report, in a synthetic sweep, that their method “actually outperforms the original clustering starting from degree 50.” That single sentence names a boundary, and, as far as we can determine, no subsequent work returns to it.

What propagated was the headline rather than its scope. “Sparsify before you detect” became a general recommendation: applied to modularity optimizers rather

than cut partitioners, to sparse graphs rather than dense ones, and to graph-internal objectives rather than external references. None of these extensions is unreasonable on its face, and each was made in good faith. But together they moved the claim off the territory where it was established, and the degree threshold that would have marked the move was not carried along.

One methodological requirement was lost with it, and it is the reason the boundary has stayed invisible. A sparsifier for community detection is built to remove edges that lie between communities in preference to edges that lie inside them. A graph so treated is therefore a graph on which any partition looks better: the very edges that penalize a partition have been preferentially deleted. Measuring the quality of a partition on the sparsified graph does not measure how well sparsification preserved community structure; it measures how thoroughly the sparsifier did its own job. The correct comparison scores the partition obtained from the sparsified graph and the partition obtained from the original graph on the same graph—the original one—so that a single yardstick is applied to both.

This requirement is not our invention. It is stated in the 2011 paper, whose authors write that they “cannot simply measure the conductance using the very same sparsified graph, since that would tell us nothing about how well the sparsified graph retained the cluster structure in the original graph,” and who report every quality figure on the original graph accordingly. In much of what followed, and in the released evaluation code of at least one widely used sparsification benchmark, the requirement is absent.

Its absence is not a minor bias. Across 63 controlled configurations spanning four detection algorithms, two sparsifier families and seven networks, scoring on the sparsified graph rather than the original reverses the sign of the conclusion in 45 of them. A pipeline that is harmful by 0.40 modularity under the correct measurement can appear beneficial by 0.39 under the incorrect one. A literature that measures this way cannot locate a boundary, because on the wrong side of the yardstick everything succeeds.

We therefore rebuilt the evaluation before re-asking the question. Partitions are always scored on the original graph. Comparisons against a stochastic baseline are given the same wall-clock budget as the pipeline they are compared with, and are also checked against a fixed-restart baseline, because a runtime-matched control is weak precisely when the pipeline is fast. Because sparsification fragments partitions, and because best-match and

information-theoretic recovery scores rise mechanically with the number of clusters, every recovery comparison is matched for granularity against a partition of the original graph at the same cluster count and is referenced to a chance floor computed from size-matched random partitions. Because a heavy-tailed degree sequence alone can produce several of the effects that are attributed to community structure, the central mechanistic claims are checked against degree-preserving randomizations of the same graphs.

With that instrument we ran 23 controlled experiments over both sparsifier families that have ever claimed quality gains—degree-based and similarity-based, the latter being the 2011 method itself—four detection algorithms drawn from two families, seventeen real networks of up to 25 million edges, and a synthetic sweep of the one variable the original authors identified as decisive. The design is intended to locate a boundary rather than to confirm or refute a claim, and it is reported as such: predictions and stopping rules were fixed before each experiment ran, and several of our own hypotheses did not survive them.

The boundary exists, and it is sharp.

Below it, sparsification does not help. For modularity-family detectors with free granularity, on the sparse graphs that dominate practice—average degrees between 5 and 33—no sparsifier we tested improves modularity measured on the original graph beyond the variation of a few random restarts, at any retention level that preserves quality, under any of four algorithms. Nor is there a speed argument: end-to-end, including the sparsifier’s own cost, the pipeline runs at between $0.87\times$ and $1.35\times$ the time of clustering the original graph directly, and at aggressive retention it can be slower, because a fragmented graph gives the optimizer more work rather than less. Every apparent gain we could find in this regime—in modularity, in ground-truth recovery, and in the one case where a sparsifier’s edge-selection signal is demonstrably structure-aware rather than degree-driven—dissolves when granularity is matched, when chance is subtracted, or when the baseline is given an equal compute budget.

Above it, sparsification helps, and it helps for the reasons the 2011 authors gave. On synthetic graphs with planted communities, holding the number of communities fixed by construction so that no granularity effect is possible, similarity-based sparsification raises both the objective measured on the original graph and chance-corrected agreement with the planted labels once the average degree passes approximately 50. The effect is absent at average degree 11 and 25, present in two-thirds of configurations at 49, and present in all of them at 102 and 221; it survives the worst case over ten independent runs of the partitioner; and it is not reproduced by degree-based or uniform sparsification at identical retention, so it is a property of the similarity signal rather than of the edge budget. Injecting spurious edges makes the gain grow monotonically, which is the signature the denoising explanation predicts.

What decides the outcome, then, is not which sparsifier

is used. It is the average degree of the graph, the family of the detection algorithm—fixed- k balanced partitioners gain where free-granularity modularity optimizers do not—and whether quality is measured against an external reference or against an objective defined on the graph whose edges were deleted. The mechanism is consistent with this: the quantity that governs a degree-based sparsifier’s effect on modularity is the separation between the scores it assigns to within-community and between-community edges, and we show by direct intervention that this quantity is controlled by where high-degree-product edges sit relative to community boundaries, at fixed degree sequence, fixed partition and fixed modularity.

Two facts must travel with the positive result. Sparsification repairs a weaker detector; it does not overtake a stronger one, and a resolution-tuned modularity optimizer on the untouched graph outperforms every sparsified pipeline we measured. And with an exact similarity computation the quality gain is not a speed gain, because the sparsifier costs more than the detection it accelerates.

This paper contributes the following.

- The boundary. A degree threshold near average degree 50 and an algorithm-family split, established under controls that exclude granularity by construction, together with the negative territory delimited across two sparsifier families, four detection algorithms and seventeen networks.
- An evaluation protocol, and the measured cost of omitting it. Each control is motivated by a specific failure it catches, and the consequence of the central omission is quantified: 45 sign reversals in 63 configurations. We also introduce a permuted-weight control for similarity-weighted pipelines, which shows that the benefit attributed to similarity weighting is reproduced, and in one case exceeded, when the weights are randomly permuted across edges.
- A mechanism for the boundary. A decomposition of the score-separation quantity into a sorting term and a granularity term across seventeen networks, and a direct causal test in which that quantity is steered through more than a full sign change while degree sequence, partition, intra-edge fraction and modularity are all held exactly fixed.
- An open phenomenon. In several controlled settings the partitions that best recover reference communities are not the partitions that score best on the objective, and the two criteria move in opposite directions within the same configuration. We report this as an observation rather than a result, and argue that it is the most consequential open question the boundary exposes.

II. Background and Setting
III. What Honest Evaluation Requires
IV. The Mechanism
V. Where Sparsification Does Not Help
VI. Where It Does
VII. Discussion
VIII. Conclusion
References

- [1] V. Satuluri, S. Parthasarathy, and Y. Ruan, “Local graph sparsification for scalable clustering,” in Proceedings of the 2011 ACM SIGMOD International Conference on Management of Data, 2011, pp. 721–732.